A robot that acts to know

Day 2 — The CORTEX architecture: concepts and memories

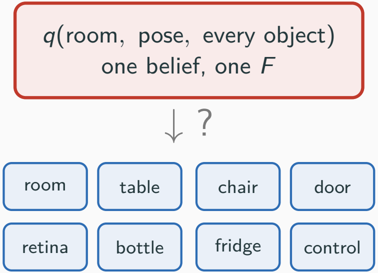
Course built on the doctoral thesis of Noé Zapata

Tuesday 1 December

Montevideo, 30 November – 4 December 2026

Who holds the belief?

What yesterday's equation did not say


q(room, pose, every object)
one belief, one F
?
room
table
chair
door
retina
bottle
fridge
control

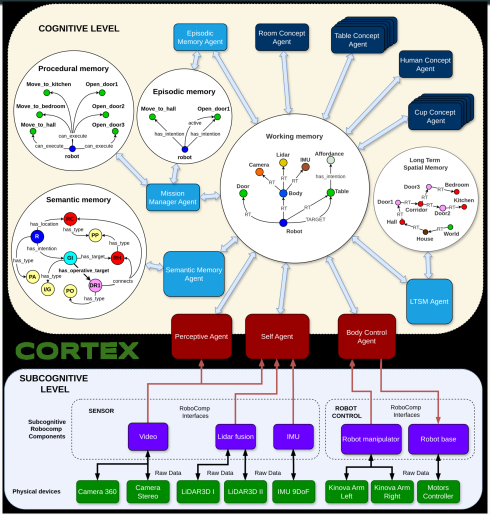
A dozen processes, sensors at 30 Hz, a robot in motion.

- **Where** is the objective computed?
- **Who** holds the beliefs?
- What happens when they must **agree** on what the world contains?

The answer is an architecture, **CORTEX**. Its fabric is **concepts**, and today's claim is that it is a factorisation of q.

source: cortex.tex l.107–109, l.377–378

CORTEX


COGNITIVE LEVEL
Episodic Memory Agent
Room Concept Agent
Table Concept Agent
Human Concept Agent
Cup Concept Agent
Procedural memory
Move_to_kitchen
Open_door1
Move_to_bedroom
Open_door2
Move_to_hall
Open_door3
can_execute
robot
Episodic memory
Move_to_hall
Open_door1
active
has_intention
robot
Working memory
Lidar
Camera
IMU
Affordance
has_intention
Door
Body
Table
RT
TARGET
Robot
Long Term Spatial Memory
Door3
Bedroom
Door1
Corridor
Kitchen
Door2
Hall
House
World
Mission Manager Agent
Semantic memory
has_location
has_type
has_intention
has_target
has_operative_target
connects
RC
R
PP
GI
RH
PA
I/G
PO
DR1
Semantic Memory Agent
LTSM Agent
CORTEX
Perceptive Agent
Self Agent
Body Control Agent
SUBCOGNITIVE LEVEL
SENSOR
RoboComp Interfaces
ROBOT CONTROL
RoboComp Interfaces
Subcognitive Robocomp Components
Video
Lidar fusion
IMU
Robot manipulator
Robot base
Raw Data
Physical devices
Camera 360
Camera Stereo
LiDAR3D I
LiDAR3D II
IMU 9DoF
Kinova Arm Left
Kinova Arm Right
Motors Controller

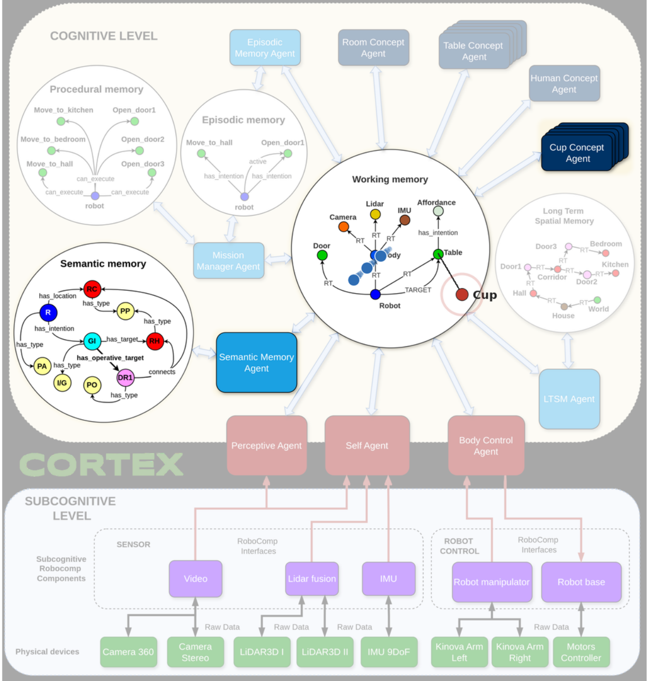
A cognitive architecture for autonomous robots: memories as graphs, and agents attached to them.

- 🟦 concept agents: one class, one belief
- 🟦 agents that manage the memories
- 🟥 agents that reach the body
- 🟪 RoboComp components

At the centre, the **working memory**: the current state, as one distributed graph.

This is the design: the semantic and procedural circles are not built yet.

source: cortex.tex l.110–113, l.154–170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CORTEX in action

1. Sensors fire
2. Two facts land on the graph
3. Concept agents read it
4. They revise their beliefs

5. Something new is born

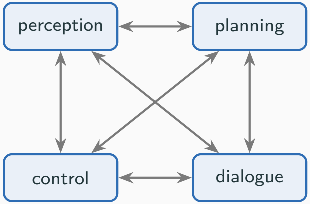
A cup node appears: a structural change. Semantic memory checks it makes sense.

6. A mission is supplied
7. An object makes an offer
8. The controller decides
9. Episodic memory keeps the record
10. The robot leaves the room

🟥 sensor data ● beliefs ◆ offers, actions

Schematic. On the robot all of this runs at once, at different rates.

Modules that talk, or agents that share



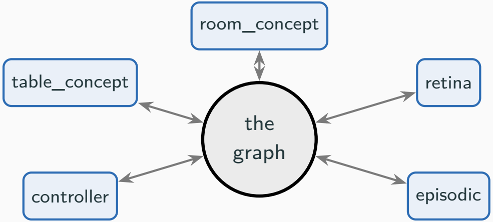
perception

planning

control

dialogue

$n(n-1)/2$ interfaces



room_concept

table_concept

retina

the graph

controller

episodic

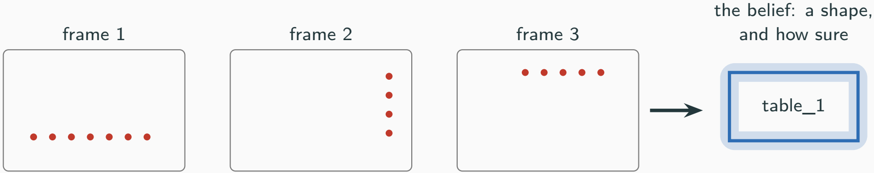
n connections

There are no agent-to-agent messages. The count is the least of it: there is one place that answers “what does the robot currently believe?”

source: cortex.tex l.345–360

Concepts: the fabric of CORTEX

A concept agent is neither a detector nor a service

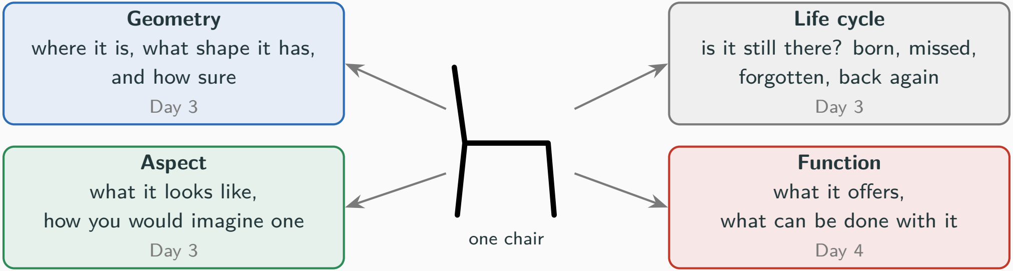

frame 1
frame 2
frame 3
the belief: a shape,
and how sure
table_1

The camera only ever delivers fragments, one face at a time, never the object.

The object is a construct the agent maintains across fragments, and each cycle's belief is the next cycle's prior.

source: concept_agents.tex l.90–95, l.163–167

A concept has dimensions



Geometry
where it is, what shape it has,
and how sure
Day 3

Aspect
what it looks like,
how you would imagine one
Day 3

one chair

Life cycle
is it still there? born, missed,
forgotten, back again
Day 3

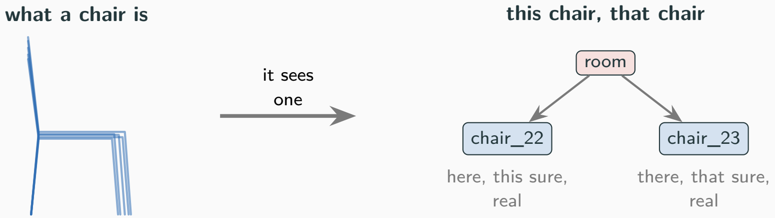
Function
what it offers,
what can be done with it
Day 4

One kind of thing in the world, one concept agent, answering for all four.

Concepts are the fabric of CORTEX: every memory below is concepts, kept for longer.

source: the four dimensions are this course's way of laying out the thesis

The agent holds the class. The graph holds the instances.


what a chair is
it sees
one
this chair, that chair
room
chair_22
chair_23
here, this sure,
real
there, that sure,
real

inside chair_concept: the class

on the graph: the instances

The class is the prior and stays in the agent. **An instance is a posterior** and is published.

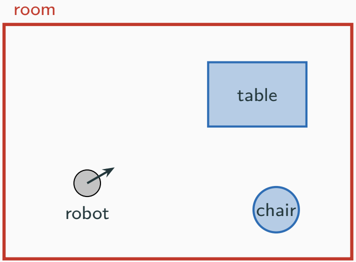
Asked “how do you imagine a chair?”, the language agent samples the class, with no data.

How much the class does depends on the object: a chair’s shape comes from the class and only its pose is inferred; a table’s shape is fitted.

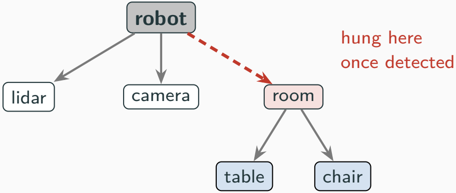
source: common/concept_manifest/chair.concept.toml; the tool `imagine_view`

The shared graph

The working memory: a scene, written as a graph


room
table
robot
chair

what is out there


robot
lidar
camera
room
table
chair

hung here once detected

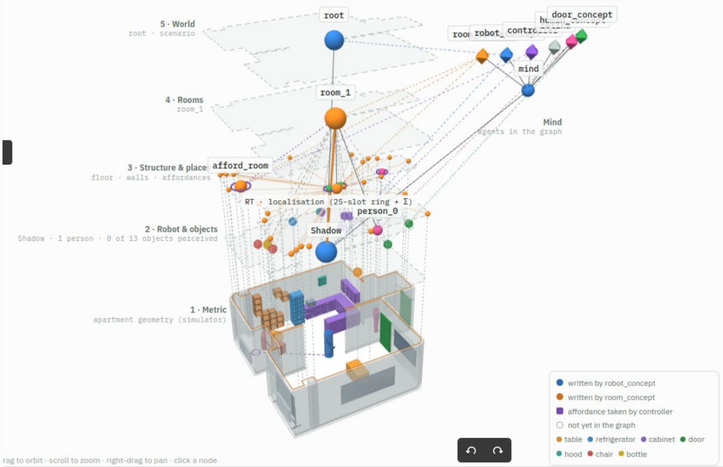
what the robot believes

The robot is at the centre of everything. Once the room is detected, it is hung from the robot, and the objects from the room. Each is an instance, owned by its concept agent.

Where is the table, seen from the lidar? Name two nodes and ask the graph.

source: cortex.tex l.231–246

The whole graph, from a simulated run


5 · World
root · scenario
root
door_concept
hu
roo
robot_
contro
mind
Mind
agents in the graph
4 · Rooms
room_1
room_1
3 · Structure & place
floor · walls · affordances
afford_room
RT · localisation (25-slot ring + Σ)
person_0
2 · Robot & objects
Shadow · 1 person · 0 of 13 objects perceived
Shadow
1 · Metric
apartment geometry (simulator)
● written by robot_concept
● written by room_concept
■ affordance taken by controller
○ not yet in the graph
● table ● refrigerator ● cabinet ● door
● hood ● chair ● bottle
rag to orbit · scroll to zoom · right-drag to pan · click a node

Layered by what a node is about: robot and objects, structure and places, rooms, world.

Colour says **who wrote it**.

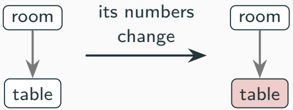
Top right, **Mind**: the agents are nodes too.

The bottom layer is the simulator's apartment, not a belief.

source: the 3D graph viewer

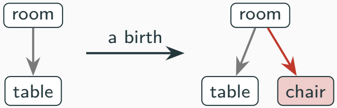
Two kinds of change

Something known has moved


room
table
its numbers change
room
table

The belief is revised by data.

The world contains something new

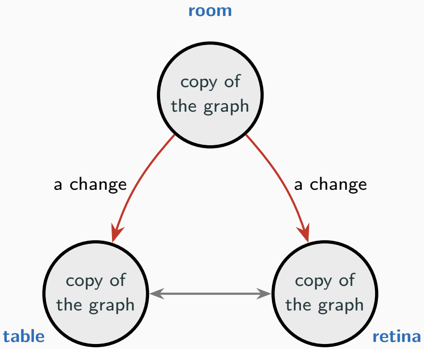

room
table
a birth
room
table
chair

The structure of the belief itself changes.

Agents watch for one kind or the other. Tomorrow is about the second.

source: cortex.tex l.248–253, l.659–671

Everyone holds a copy, and sends corrections

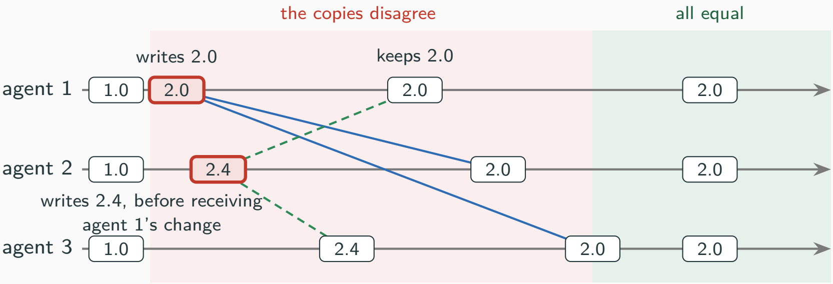

room
copy of the graph
a change
a change
copy of the graph
copy of the graph
table
retina

No central server, and nobody waits for a lock. Changes merge by themselves. Two consequences:

- Changes are signed by their writer, so two agents with one name corrupt the graph.
- Your copy may be a moment behind mine. Nothing may assume we see the same instant.

source: cortex.tex l.255–272

Copies may disagree for a moment, and then they agree

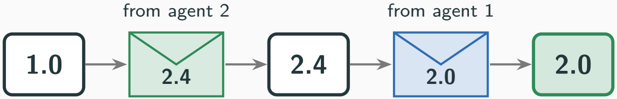

the copies disagree
all equal
writes 2.0
keeps 2.0
agent 1
1.0
2.0
2.0
2.0
agent 2
1.0
2.4
2.0
2.0
writes 2.4, before receiving agent 1's change
agent 3
1.0
2.4
2.0
2.0

Nobody waited, and nobody locked. For a while agent 3 believes 2.4. "Eventually consistent" means only that, once the messages have arrived, every copy holds the same value.

source: cortex.tex l.255–272

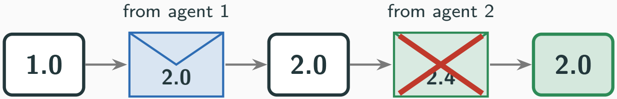
Same letters, any order, same result

One copy receives agent 2's letter first



1 outranks 2: it replaces

Another copy receives agent 1's letter first



2 does not outrank 1: dropped

Each wrote before receiving the other's change, so a fixed rule decides, the same in every copy: **the lower id wins**. A writer that had already received the other's change simply replaces it.

But the ids are arbitrary!

	three copies		
ids as given	2.0	2.0	2.0
ids swapped	2.4	2.4	2.4

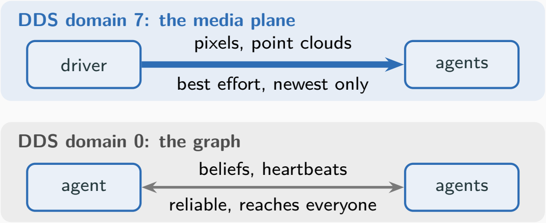
The id picks *who* wins, never *whether* the copies agree: every copy ranks the same two letters the same way.

The rule buys agreement, not the better value.

One of the two writes is always lost. That is why **each node has a single owner**: with one writer, there is nothing to collide with.

source: cortex.tex l.255–272; cortex, core/crdt/delta_crdt.h

Sensor data travels beside the graph



DDS domain 7: the media plane

pixels, point clouds

driver

agents

best effort, newest only

DDS domain 0: the graph

beliefs, heartbeats

agent

agents

reliable, reaches everyone

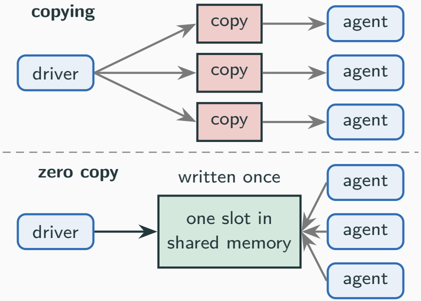
A **DDS domain** is a private channel: processes in different domains never discover each other.

Images written into the graph starve it, and the first symptom is a missed heartbeat: an agent declared dead because a camera was busy.

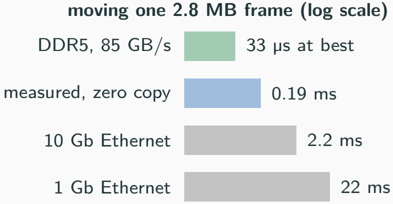
The graph holds a sensor's *existence*, and where to find its stream.

source: cortex.tex l.375–381, l.386–402, l.410–415

On one computer: pass the place, not the image


copying
driver
copy
copy
copy
agent
agent
agent
zero copy
written once
driver
one slot in shared memory
agent
agent
agent

Readers look at the same bytes in place. The cost does not grow with readers.

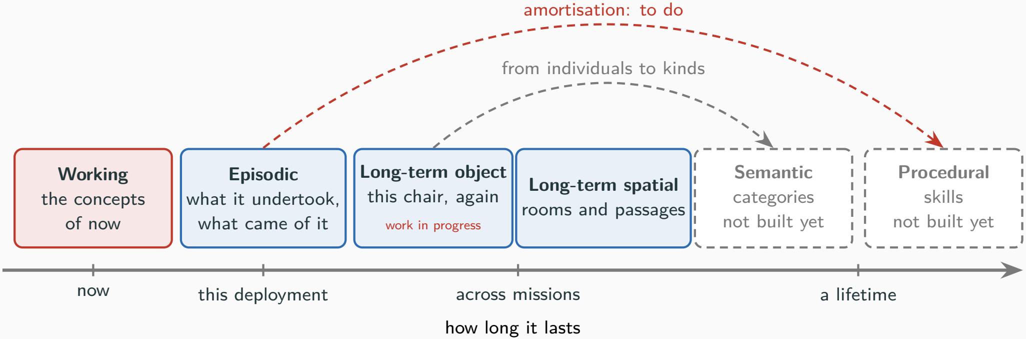


DDR5 typically moves 80–90 GB/s. Two cameras at 30 Hz are only about 0.2 GB/s. Most of the 0.19 ms is not moving bytes.

source: cortex.tex l.400–406, l.432–435; arithmetic: 2.8 MB at each rate

Memories

Memories: the same concepts, kept for longer

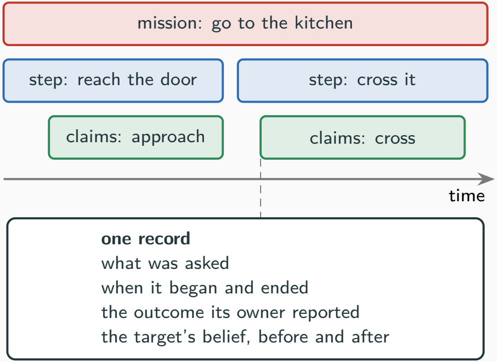

amortisation: to do
from individuals to kinds
Working
the concepts
of now
Episodic
what it undertook,
what came of it
Long-term object
this chair, again
work in progress
Long-term spatial
rooms and passages
Semantic
categories
not built yet
Procedural
skills
not built yet
now
this deployment
across missions
a lifetime
how long it lasts

Each memory has an agent that manages it, and fills it by watching the working memory.

Agents still only talk through memories. For now, the classes stay inside the concept agents.

source: cortex.tex l.519–537, as built on 2026-11-09

Episodic memory: what it undertook, and what came of it


mission: go to the kitchen
step: reach the door
step: cross it
claims: approach
claims: cross
time

one record
what was asked
when it began and ended
the outcome its owner reported
the target's belief, before and after

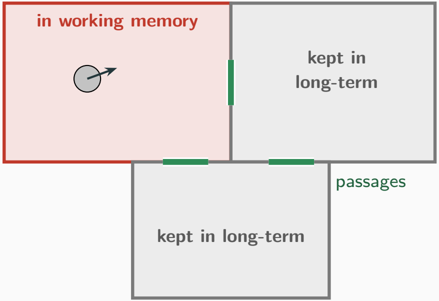
Its unit is one thing the robot undertook to do, at any level.

The start and the end are read from the controller's journal, not guessed from the world.

At the turn of the day, records fold into **track records**: how often crossing this door has worked, and how long it takes.

source: cortex.tex l.539–547; the example mission is illustrative

Long-term spatial memory: rooms and passages


in working memory
kept in long-term
passages
kept in long-term

The room the robot leaves moves out of working memory, and goes back in when the robot returns.

Working memory stays small, and the map is bounded only by storage.

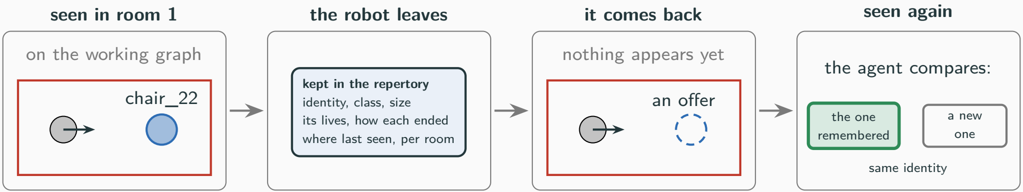
It plans routes across rooms and predicts a room before entering it.

It holds rooms and passages only. Objects now have a memory of their own.

source: cortex.tex l.558–564; ltsm_agent on main, a2baaebe (the objects moved out after the thesis text)

Long-term object memory: this chair, again

work in progress



seen in room 1

on the working graph

chair_22

the robot leaves

kept in the repertory
identity, class, size
its lives, how each ended
where last seen, per room

it comes back

nothing appears yet

an offer

seen again

the agent compares:

the one remembered

a new one

same identity

The memory only **offers**. The concept agent **decides**, and the object appears only once it is detected.

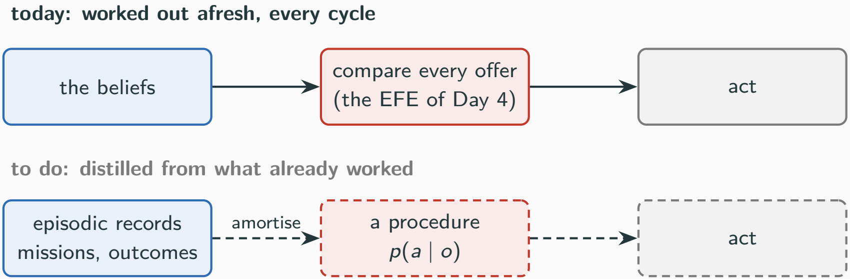
Today the comparison uses position alone. The chair's look is the missing term.

Measured: chairs only, so far

Room 1, room 2, room 1: two chairs came back with their own identities, $p = .945$ and .969.

source: ltom_agent/README.md; run ltom_chair2, 2026-11-08

Still to build: amortisation



today: worked out afresh, every cycle

the beliefs

compare every offer (the EFE of Day 4)

act

to do: distilled from what already worked

episodic records missions, outcomes

amortise

a procedure $p(a \mid o)$

act

Missions run today, and their results are kept. Nothing generalises from them yet.

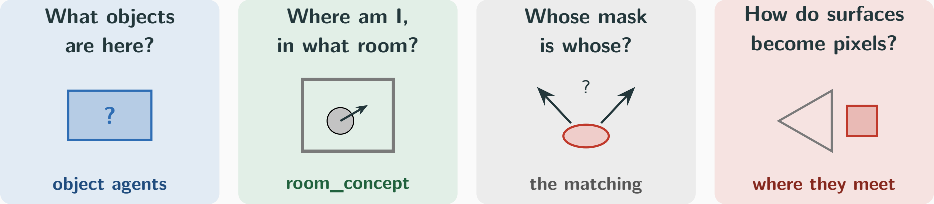
$p(a \mid o)$ is the factor from Day 1 that control and RL learn directly.

The open question: does a learned habit know when it no longer applies?

source: a plan, not a result; Day 1, "Three things you can build from one joint"

One free energy, many owners

One big belief about everything



What objects are here?

object agents

Where am I, in what room?

room_concept

Whose mask is whose?

the matching

How do surfaces become pixels?

where they meet

Before any data, the four are independent: objects do not know where the robot is.

Once data arrive they are tangled: the scan that locates the wall also hits the refrigerator standing against it.

source: concept_agents.tex l.255–293

Not everything unknown is worth reporting

Answer it

the table's shape,
whether it is real

published on the graph,
with how sure

Allow for it

the robot's exact path,
the exact room

needed to read the data,
its doubt is carried along

Just guess it

which mask belongs
to which object

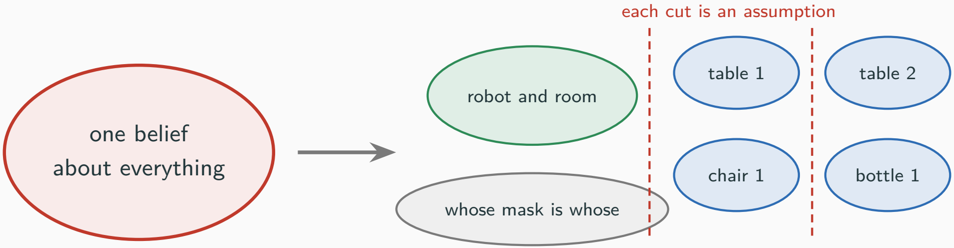
one best guess, because
trying all is impossible

The third has no principled defence, and the thesis says so.

A SLAM system would also *report* the path. That one change is what separates this design from a monolithic one.

source: concept_agents.tex l.395–420

Cut the belief into pieces, one process per piece

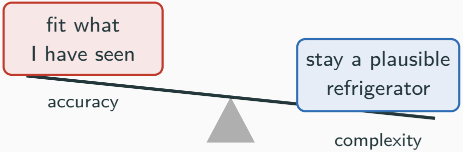

each cut is an assumption
one belief
about everything
robot and room
whose mask is whose
table 1
table 2
chair 1
bottle 1

Across a cut, two beliefs no longer inform each other, except through what is published.

A process boundary is a statement about the world, not only about the software.

source: concept_agents.tex l.114–123, l.447–455

What each agent minimises: yesterday's trade, once per object


fit what I have seen

stay a plausible refrigerator

Exactly Day 1's F, for one instance:

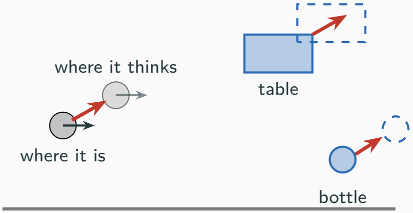
$$F_i = \text{misfit} + \text{distance from the class}$$

A refrigerator is about 0.60 ± 0.08 m deep before it is ever seen: the class does a lot of the work.

The agent takes the robot's published pose as given, and classes never add their F together.

source: concept_agents.tex l.347–351, l.496–516

The smallest scene that shows the price



the same arrow, for everything in this frame

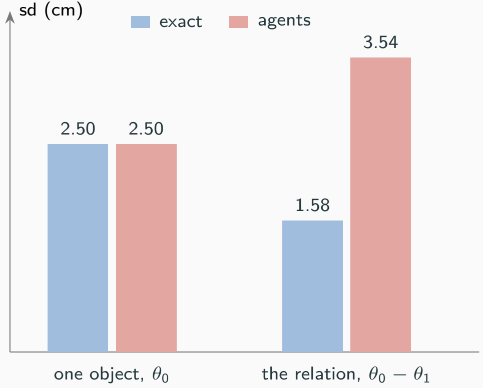
The robot places what it sees using its own pose. If the pose is off, every object in that frame is off by the same amount:

$$z_{it} = \underbrace{\theta_i}_{\text{true place}} + \underbrace{\delta_t}_{\text{shared}} + \underbrace{\varepsilon_{it}}_{\text{its own}}$$

Without a shared error, the sum of the agents' F_i is the global F. With one, dividing the work is an assumption.

source: lab 2, part C; the shared offset: concept_agents.tex l.64–70

Each belief is right. The relation between them is wrong.

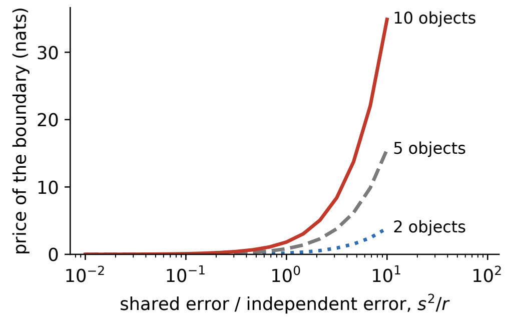


The shared error cancels in a difference. So the robot knows how far the bottle is from the table far better than it knows where either is.

Two independent beliefs on a graph cannot say that. The price is paid in relations.

source: lab 2, exercise C2 ($r = (5\,\mathrm{cm})^2$, $s^2 = (10\,\mathrm{cm})^2$, 20 frames, 3 objects)

The price of a boundary, in nats



How far the agents' beliefs are from the exact one, in Day 1's units:

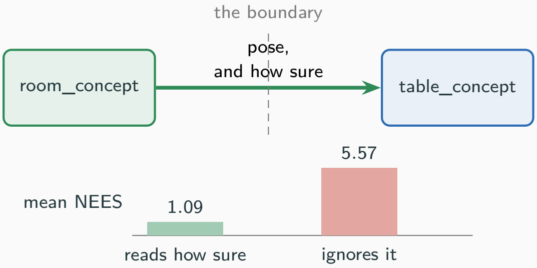
$$F[\text{agents}] - F[\text{exact}]$$

- grows with shared error and with the number of objects;
- does not fall with more data: 2.56 nats at 10, 40 and 160 frames.

Data sharpen each factor. They cannot restore a correlation the family has no place to hold.

source: lab 2, exercise C3

A mean is half a belief



The NEES compares a belief's real errors with the errors it claims: **1 is honest**.

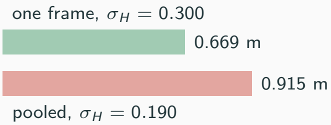
An agent that reads *how sure* the robot is stays honest. One that takes the pose as truth claims five times too small a variance.

The covariance travels with the mean, so any process can use the estimate as evidence and not as ground truth.

source: lab 2, exercise B2; concept_agents.tex l.90–95

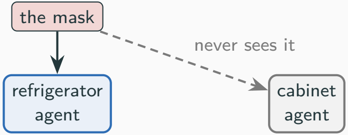
The architecture carries its failures on its face

Pooling eight frames into one



Each frame brings its own offset, the fit has nowhere to put eight of them, so they land in the shape. More confident about a worse answer.

A cabinet labelled as a fridge

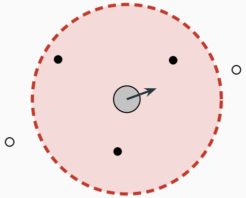

the mask
never sees it
refrigerator agent
cabinet agent

The comparison that would settle it, how well each class explains the same pixels, is never made.

source: concept_agents.tex l.73–85; agents_life_cycle.tex l.808–809

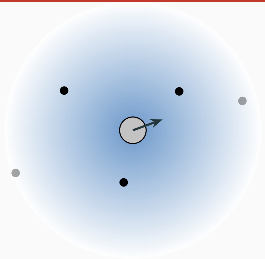
No thresholds

A gate is what a model looks like before the modelling is done



a gate

"ignore anything beyond 5 m"



a precision

every sighting counts, as much as it can be trusted

far away: less sure about direction

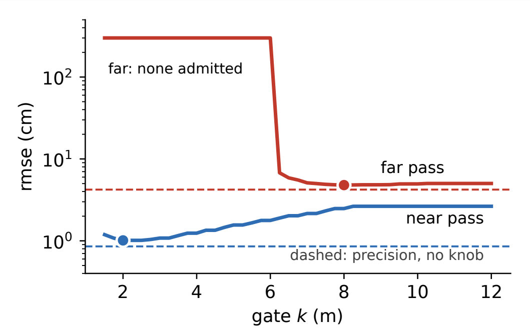
moving: the image is blurred, so a larger variance

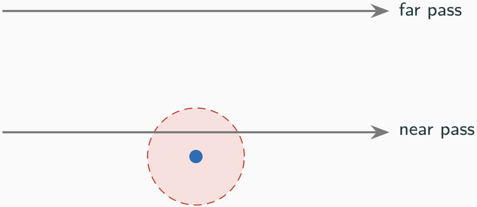
many points: they share one error, so they stop helping

No rule on top of the inference: the behaviour falls out of the model.

source: concept_agents.tex l.552–562

A gate is a parameter the scene can invalidate



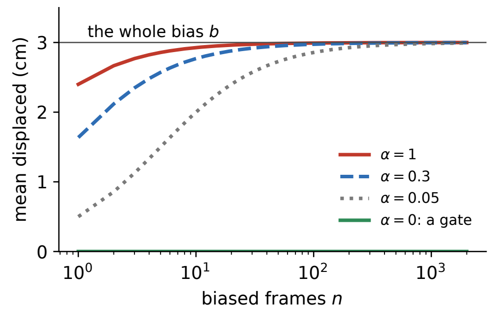


Tuned on the near pass (2 m), the gate admits nothing on the far one: 3 m of error.

The precision needs no tuning and wins on both: 0.86 against 1.01 cm, 4.22 against 4.79 cm, each against the *best* gate for that pass.

source: lab 2, part D

The gate that survives



a run of frames, all wrong the same way

Turning their weight α down only slows it. In the end the estimate is off by the whole bias. Only ignoring those frames stops it.

A variance handles errors that average. This one does not average.

source: agents_life_cycle.tex l.783–791; lab 2, part E

Lab

Lab 2: agents on a shared graph

`labs/day2.ipynb`, about two hours.

1. **The graph**: an ownership contract, and a sweep for the nodes of dead processes.
2. **Agents that only read and write**: an object agent that reads the robot's belief from the edge. Then remove the covariance and measure the NEES.
3. **One F, many owners**: when the sum of F_i is the global F; the marginal against the relation; the price of the boundary in nats.
4. **No thresholds**: a range gate against a precision, on two passes.
5. **The gate that survives**: a bias no precision can bound.

Tomorrow

Today a concept was one box. Tomorrow we open it:

- **geometry**: the room, the pose, the objects;
- **life cycle**: birth as model comparison, absence as evidence, forgetting as inferred volatility;
- **aspect**: what a thing looks like, and why it matters on a second sight.

Reading for Day 3

Thesis chapters:

- *The Life Cycle of a Concept Agent*;
- *The Room Concept Agent and Robot Localisation*.

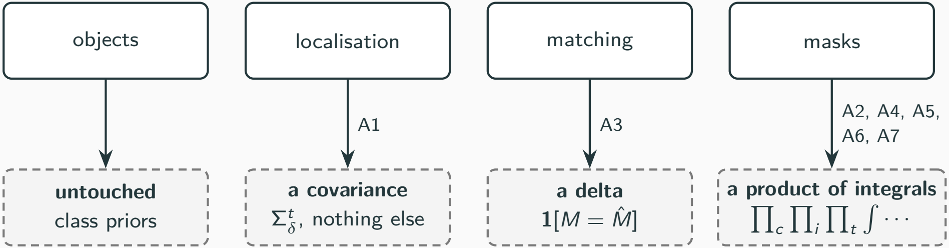
Backup: the whole scene, in causal order

objects exist and have shape	the room has walls, the robot moves, wheels report it, LiDAR hits all of it	what is visible from where	surfaces become masks
$P(\ell)\,P(\theta\mid\ell)$	$P(\mathcal{R})\,P(X)\,P(\Phi\mid X)\,P(L\mid X,\mathcal{R},\theta)$	$P(M\mid\theta,\ell,X)$	$P(Z\mid\theta,M,X)$
object agents	**room_concept**	**the matching**	**where they meet**

- Objects and pose are independent *a priori*: objects do not know where the robot is.
- Once data arrive they are coupled twice: through the masks, and through the LiDAR sweep, which hits the fridge as well as the wall behind it.

source: concept_agents.tex l.255–293

Backup: from the exact posterior to what actually runs



objects → **untouched** class priors

localisation —A1→ **a covariance** Σ_{δ}^{t}, nothing else

matching —A3→ **a delta** $\mathbf{1}[M = \hat{M}]$

masks —A2, A4, A5, A6, A7→ **a product of integrals** $\prod_c \prod_i \prod_t \int \cdots$

Seven assumptions, stated one at a time, each with its justification and its cost:

A1 objects as landmarks · A2 pose error integrates per frame · A3 hard matching · A4 classes apart · A5 instances apart · A6 geometry / existence · A7 Markov time

source: concept_agents.tex l.568–613, l.626–1019

Backup: the factorised belief, and one agent's objective

$$q(\theta, \ell, M, X, \mathcal{R}) = \underbrace{q(X, \mathcal{R})}_{\text{foreign}} \; \underbrace{\mathbf{1}[M = \hat{M}]}_{\text{degenerate}} \; \prod_{c} \prod_{i} \underbrace{q(\theta_i)}_{\mathcal{N}(\mu_i, \Sigma_i)} \; \underbrace{q(\ell_i)}_{\text{log-odds}}$$

$$F_i = \underbrace{\mathbb{E}_q\big[-\ln P(\bar{Z}_i \mid \theta_i, \hat{X})\big]}_{\text{accuracy}} + \underbrace{D_{\mathrm{KL}}[q(\theta_i) \,\|\, P(\theta_i \mid \ell_i)]}_{\text{complexity}}$$

The localisation block is held fixed; the matching contributes nothing to F.

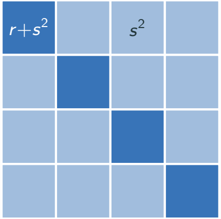
source: concept_agents.tex l.447–455, l.468–516

Backup: the shared error, as a matrix

Integrate δ_t out of $z_{it} = \theta_i + \delta_t + \varepsilon_{it}$, and the noise of the n sightings of one frame has covariance

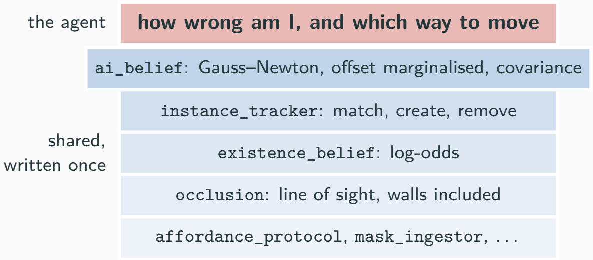
$$C = r\,I + s^2\,\mathbf{1}\mathbf{1}^\top.$$

The agents use only its diagonal, $R = r + s^2$.



source: lab 2, part C

Backup: An agent writes two things



the agent

how wrong am I, and which way to move

`ai_belief`: Gauss–Newton, offset marginalised, covariance

`instance_tracker`: match, create, remove

shared,
written once

`existence_belief`: log-odds

`occlusion`: line of sight, walls included

`affordance_protocol`, `mask_ingestor`, . . .

A residual and its Jacobian. All the Bayesian mathematics lives in the engine, once.

This is why a roster entry can be half a page: bottle needs little, table and refrigerator fill the template, and that is a fact about the objects.

source: concept_agents.tex l.534–537, l.136–138, l.1196–1203

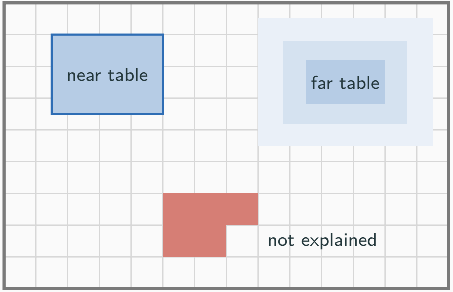
Backup: Each publishes what only it can compute

computes	writes on the graph	reads it
room_concept	where the robot is, and how sure	object agents
an object agent	its shape, and how sure	planner and residual
an object agent	whether it is real; what it offers	the controller
residual_concept	occupied and *not explained*	planner

Nobody asks anybody. Everything an agent knows about another's work is on the graph.

source: concept_agents.tex l.1209–1216, l.1533–1544

Backup: residual_concept: whatever nobody else explains



a sure table claims its cells tightly; an unsure one loosely

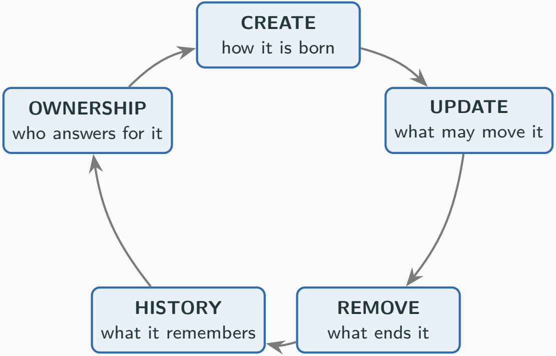
A grid of what the sensors hit, minus what every other agent claims to explain. What is left is what the robot must avoid.

Nothing has to be recognised for the robot not to hit it.

No safety margin anywhere: each claim is as wide as the claimant's own published uncertainty. That is why the others must publish it.

source: concept_agents.tex l.1497–1538

Backup: Every agent answers to a common contract



CREATE
how it is born

UPDATE
what may move it

REMOVE
what ends it

HISTORY
what it remembers

OWNERSHIP
who answers for it

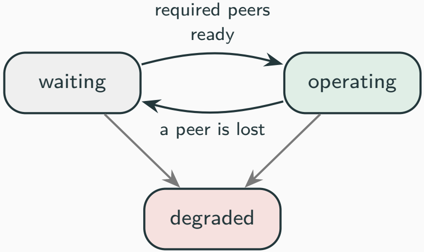
Not paperwork: this is what anything that integrates noisy fragments has to do.

A rule written in a contract is a rule a new agent cannot forget.

Tomorrow: where beliefs come from and how they end.

source: concept_agents.tex l.169–180; CONCEPT_AGENT_LIFECYCLE.md l.7–13

Backup: The factorisation is only true while its owners are alive


required peers
ready
waiting
operating
a peer is lost
degraded

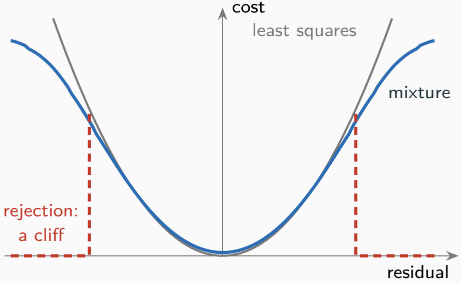
Specified in TLA$^+$ and model-checked: 8 safety and 4 liveness properties, no counterexample.

A process crashes and its node stays. It looks like any other data, and nobody is keeping it true any more.

A peer that is present is *running*, which is not the same as *able to serve*. Heartbeats are ordinary graph nodes.

source: cortex.tex l.732–741, l.757–775, l.780–785

Backup: An outlier costs a bounded amount, and nobody rejects it



Each point has three possible explanations: the object, clutter, or a known wall.

A point that fits nothing costs a finite amount and stops pulling on the estimate.

Nobody decided to reject it. The bound is a probability in the model.

source: concept_agents.tex l.518–524

Backup: The rule is: declare it, justify it, or remove it

ignoring a bad frame: stays
no precision can bound that bias

birth after N sightings: must go
a count standing in for nats

waiting a few seconds for a peer: a debt
whether a peer is alive is a belief too

Measured: a test that graded itself

A chair's "can I see its yaw?" was computed from its own yaw. The yaw was about 45° wrong, so the test said no, and the wrong yaw froze.

source: agents_life_cycle.tex l.1276–1282; cortex.tex l.796–804; CONCEPT_AGENT_LIFECYCLE.md l.75–79